



Freedom of speech or freedom of reach? Strategies for mitigating malicious content in social networks

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ABSTRACT

Malicious content threatens the integrity and quality of content in social networks. Research and practice have experimented with network intervention strategies to curb malicious content propagation. These strategies lack efficiency, target malicious content propagators, and abridge freedom of speech. We draw upon the preferential attachment literature and cognitive load theory to employ the mechanisms of network formation, information sharing, and limited human cognitive capacities to propose an alternative feed management strategy—Preferentiality Dampened Feed Management. We compare and contrast this strategy against other established strategies using an agent-based model that utilizes empirical data from Twitter and findings from the prior literature. The results from our two experiments suggest that our proposed strategy is more effective in curbing malicious content propagation than other established strategies. Our work has important implications for the network interventions literature and practical implications for platform providers, social media users, and society.

1. Introduction

Malicious content spreads virally on social networks and has real-world consequences [64]. Malicious content diffusion on social networks has led to the misallocation of resources during natural disasters, promoted health-related misinformation, sowed distrust, manipulated individuals' beliefs, and swayed political opinions and support [17,64]. Malicious content is often intentionally created [64]. Such malicious content may encompass but is not limited to, fake news and other deviant content such as fraudulent messaging and online trolling [62]. This emergence and diffusion of intentional malicious content has substantially threatened the integrity and quality of content on social networks [64]. A recent economic study revealed that fake news costs the global economy \$78 billion yearly [5]. Therefore, misinformation detection [e.g., [7,16,19,67]] and the development of strategies that can help effectively curb the spread of malicious content [14,24] have been in the focus of researchers and practitioners alike.

Researchers have proposed several strategies to counteract the

spread of malicious content [1,24,62]. Some of the most popular strategies involve network alterations that target individual users or specific segments of the social network [62]. For example, Valente [62] and Harris et al. [24] investigate the influence of network alteration strategies on social networks that involve the removal of malicious agents (e.g., bots) and/or content from social networks. Such practices are commonly referred to as bot removal [14]. Past research has also deployed persuasion strategies. For example, researchers have recommended nudging users or providing information external to a user's networks to promote positive behaviors Aral [1]. Similarly, several other researchers (e.g., [12,20,49,57]) argue for fighting the spread of misinformation on social networks by promoting fake news alerts or building fact-checking platforms. Social networks have employed different strategies in practice. For example, Twitter implemented a network alteration strategy and removed around 70 million suspicious accounts from their network in 2018. Since then, popular social networks have followed such network alteration strategies.

However, network alteration strategies have been noted to have

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limited effects and come at a significant cost to social network providers for the following reasons. First, removing suspected malicious accounts diminishes their user base—a key metric for measuring social network growth—thereby negatively affecting the bottom line of these social networks. Second, network alterations often have a short-lived limited impact [2,37]. The whack-a-mole nature of this approach requires social networks to remove suspected malicious accounts continuously [19,47]. Third, there is a burgeoning discussion regarding whether social network posts fall under the purview of freedom of speech [23,28]. Many have contended that removing accounts from social networks like Twitter raises claims of the platform's unchecked power and can be seen as a violation of users' freedom of speech, leading to censorship concerns [10,28]. Finally, network alteration strategies are criticized for focusing on symptoms rather than the drivers of malicious content [30]. Therefore, researchers need to propose alternate yet effective network interventions that target the primary means of content propagation to curb the spread of malicious content.

We propose an alternative strategy with the objective to tackle the mechanisms responsible for content propagation in social networks and reduce the mechanisms' impact for malicious content so that the content's exposure is limited while legitimate content remains untouched. Unlike network alterations, our strategy—Preferentiality Dampened Feed Management (PDFM)—does not censor any content from the social network. Instead, PDFM draws on preferential attachment (PA) and cognitive load as two critical mechanisms that govern the content that social network users are exposed to via their feeds. PA is dominant in forming social networks like Wikipedia and Twitter [6,53]. PA controls how new users entering a social network are more likely to link to popular users over average social network users [3]. Cognitive load also has been studied in the context of communication and interaction between users of social networks [4]. Cognitive load suggests that individuals' cognitive processing capacities are limited; they can only consume a subset of all available information – i.e., limited information processing (LIP; [25,51]). Once content is flagged as malicious by the social network providers, PDFM works toward dampening the preferential display of that content, thereby minimizing its exposure. Even though both PA and cognitive load have been extensively studied in the social network context, their impact on content propagation is often overlooked. We argue that PA not only helps in creating initial connections between users but, along with LIP, PA also plays a critical role in determining what content users get exposed to. In other words, users not only tend to form links with more popular users, but they also tend to be disproportionately exposed to the content of these popular users. Unlike the network alteration strategies that focus on removing content or accounts from social networks, PDFM deprioritizes flagged malicious content by targeting the core drivers of content propagation. Hence, PDFM represents a new class of network interventions that is a more effective means to curb the spread of malicious content.

To evaluate the effectiveness of PDFM against the established network alteration strategies (i.e., bot removal), we develop an agent-based model (ABM) mimicking a real-life social network. We use our ABM to run simulation experiments of different levels of PDFM and bot removal and their respective dampening impact on different content propagation metrics. We find that some levels of PDFM are highly effective at curbing the spread of malicious content and the resulting reduction in propagation is comparable to the ones achieved with 75% or 99% bot removal. Considering PDFM is much less cost and effort intensive than bot removal strategies, it provides an efficient alternative to network providers in their quest to curb malicious content propagation.

Our research makes three core contributions. First, we contribute to the network intervention literature by introducing PDFM. Our proposed strategy—namely, PDFM—represents a novel class of network intervention strategies that target the underlying mechanisms of malicious content propagation—i.e., PA and LIP. Specifically, rather than censoring malicious content or malicious agents, PDFM seeks to

deprioritize the flagged content, inhibiting its propagation. In doing so, we respond to the critique on the existing network intervention strategies being effortful, short-lived, restricting freedom of speech, or only applicable to human users [19,36,61] thereby expanding the current approaches of network alternation and persuasion recognized in the literature (e.g., [38,57]) and practice.

In addition, this study demonstrates how ABM can be used to study the intricate differences in the application of an intervention strategy. Using ABM we investigate the dampening effects of PDFM levels by isolating the influence of either the content selection or sorting process. Four PDFM levels—No PDFM, Low PDFM, Medium PDFM and high PDFM—are examined and compared. No PDFM relies on popularity-based (PA-based) selection and sorting; Low PDFM employs non-PA-based selection but retains PA-based sorting; and Medium PDFM combines PA-based selection with chronological sorting. In High PDFM both sorting and selection gets dampened. These distinctions shed light on how altering selection and sorting approaches impact content feed composition, exposure patterns, and the overall distribution of engagement within a social network. The interplay between content feed size and the selection/sorting strategies unveils their role in attenuating popularity-driven effects, thereby shaping content dissemination dynamics and enhancing the resilience against malicious content.

Second, we contribute to the literature on content propagation in social networks by substantiating the roles of PA and LIP in content propagation. PA has been studied extensively in network formation to understand how new entrants tend to form relationships with the most popular social network members [50,66]. This research demonstrates the influence of PA on the diffusion of content throughout social networks. Specifically, we show that network formation does not happen in a big bang but is an ongoing and evolving process such that PA emphasizes who is connected to whom and what content users are exposed to. Therefore, this research proposes and evaluates PDFM as a network intervention strategy considering PA in network formation and content propagation.

Moreover, we argue that humans' LIP capacities further exacerbate PA's influence on content diffusion. As PDFM attenuates the influence of PA on content propagation, it also incorporates the LIP capacity of humans [60]. PDFM proposes customized feed management policies that dampen the reach of flagged malicious accounts. Such dampening can limit the propagation of harmful content without controversial side effects such as censorship or removing users. Therefore, instead of targeting specific accounts and subsequently removing them, we propose a novel approach that relies on altering the content feed to limit the propagation of malicious content.

Finally, our research has practical implications for providers of social networks, users, and the general public. For providers of social networks, PDFM offers a way to inhibit malicious content more efficiently and sustainably than the widely used bot removal strategy. Bot removal is laborious and requires manual and repeated review, identification, and removal of malicious content. Users and the general public, in turn, benefit from legitimate content in social networks by being able to rely on an additional source of news for information gathering, a fundamental behavior in a democratic decision-making process. The ABM methodology used in this research presents additional contributions. By extending ABM methodology as a policy evaluation tool in social networks, we present an ABM model that researchers and providers of social networks can use to test different policies. Moreover, we highlight an innovative validation approach using real-world data and provide specific metrics for content propagation that may inform future ABM studies on social networks.

This paper proceeds as follows: We first review the relevant literature on content propagation and network interventions. Then, we introduce PDFM and the ABM of a Twitter-like social network that defines and establishes agent behaviors. Next, we validate the model, followed by experiments where we compare and contrast the efficacy of two network interventions — namely, bot removal versus PDFM. Finally, we discuss

the contributions and implications of this work.

2. Background

2.1. Content propagation in social networks

Social networks have become integral to individual and societal life [64]. Social networks, such as Twitter and Instagram, provide users with access to digital content and allow users to connect [33]. Users seek entertainment or social connections using social networks, gather information and news, or manage their professional branding [8]. As popular news media, social networks have a central role in information gathering and opinion formation processes and evidence suggests that structural properties of social networks can lead to opinion polarization as well as moderation [35,58,59].

Content propagation within social networks is complex. Scholarly literature has theorized a handful of different mechanisms and determinants that create complex patterns of information diffusion. Examining the diffusion structure of significant propagation events, Goel et al. [21] revealed that information spreads either virally (i.e., from peer to peer) or is broadcasted (i.e., from one specific source to many). Often, virality and broadcasting mechanisms complement each other [21]. Lou et al. [40] indicate that subtle and widespread infiltration of a simulated social network significantly damaged the overall information quality. They further contended that targeting influential individual human users was less effective than the infiltration approach in attaining content virality and thus manipulating a larger social network share [40]. Such viral infiltration is often achieved through coordinated diffusion events [11] which often involve so-called bots, i.e., synthetic agents that serve the purpose to reconfigure, embellish, and amplify content [56].

A rich stream of research has shown that the likelihood of information diffusion does not simply increase proportionately with more exposures [26,64]. Multiple factors, such as the similarity or the veracity of content, the popularity of a user, and the situational cognitive load, affect the likelihood that content will be shared [26,64,67]. Drawing on the social contagion theory, Hodas and Lerman [26] argue that users exposed to large amounts of information are less likely to be exposed to a specific piece of content than users exposed to a small amount of information. In doing so, Hodas and Lerman [26] note that limited cognitive capacities restrict the amount of information users can process. Conversely, researchers (e.g., [68]) investigated the relationship between parallel sets of content and the content creators' popularity. This stream argues that similar content emerging in parallel streams tends to inhibit the diffusion of the original content, thereby diminishing the creator's reach. However, content created by viral creators is likely to be further amplified if similar content is shared in parallel [68]. Finally, content's veracity is considered an essential factor for the diffusion pattern of information in social networks. Comparing the diffusion dynamics of true and false content on Twitter, Vosoughi et al. [64] found that false (but not necessarily manipulative) content spreads significantly faster, reaches more people, and is retweeted more often than authentic content.

2.2. Malicious content

Over the past twenty years, there has been an escalation in malicious activities across social networks [94]. Actors like social bots and trolls deliberately disseminating misleading information to contaminate information environments and shape public perspectives [42,54,55,56]. This dissemination of malicious content aims to influence opinions for the benefit of the instigators, as seen in instances like the Brexit referendum misinformation campaign. While malicious content is frequently associated with politics, commonly referred to as "fake news," its scope extends beyond mere falsehoods to encompass intentional manipulation of users' viewpoints [64]. This encompasses various forms of

misinformation, from online trolling to organized misinformation campaigns by entities such as governments and corporations [62]. The distinguishing factor in our research lies in the manipulative intent behind the spread of content, rather than its factual accuracy. Consequently, our focus revolves around content intentionally designed to sway opinions, irrespective of its veracity.

Prior research has investigated patterns of malicious content propagation on social networks [19,34,47,49]. Researchers have analyzed both the communication patterns initiated by social bots [13,19,55] and malicious content's specific linguistic and structural characteristics [19, 36, 50, 56, 84], leading to the development of various detection methods for malicious content [19,27] such as fake news and fake reviews [9,67,69]. A rich stream of research has investigated malicious content propagation through social bots because, as automated actors, bots purposely manipulate public opinion. For example, Lou et al. [40,55] simulated different bot strategies to investigate their effects on online information pollution using ABM. Lou et al. [40,55] found that the strategy of subtle and widespread network infiltration heavily polluted the information in the social network. Supporting these findings in their simulation, Ross et al. [55, 76] examined how social bots influence the formation of public opinion. They argued that social bots, which account for as little as two to 4 % of the population, are sufficient to sway opinion climates on social networks. These findings underline the threat malicious content poses to the quality of information in a social network and individual opinion formation.

2.3. Network interventions

Researchers in multiple fields have proposed network interventions to counteract the effects of malicious content in social networks. These network interventions aim to "generate social influence, accelerate behavior change, improve performance, and achieve desirable outcomes among individuals, communities, organizations, or populations" ([62], p. 49). This stream of research has suggested various network intervention strategies to fight malicious content [62]. On the one hand, persuasion strategies target individual users and aim to change content-sharing behaviors. Such interventions include digital nudges, fake news alerts, or fact-checking platforms encouraging individuals to verify their shared content (e.g., [12,14,20,49,57]). On the other hand, network alteration strategies aim to reconfigure network structures, for example, by removing individual users. Alteration strategies and their components have been deployed by social networks to contain negative user behaviors or to support positive behaviors among users [1], reduce astroturfing [24], and fight COVID-related misinformation [10]. For example, in 2018, Twitter removed roughly 70 million suspicious accounts from their network [15]. Similarly, in 2020, Google and Facebook removed COVID-19-related misinformation from their networks. Even though these strategies target the entire population of social network users and not only bots, the strategies are often referred to as 'bot removal' strategies.

While both strategies – persuasion and network alteration – have been tested and applied successfully, they have limitations concerning effectiveness, cost-efficiency, and long-term sustainability [37]. Persuasion strategies, for example, fake news alerts, news source evaluation flags, or nudges to induce fact-checking, have proven effective to reduce belief in political news in experiments (e.g., [12,20,49,57]). However, such persuasion strategies are only fit to influence the behavior of human agents. Bots, estimated to make up to 15% of Twitter's network, are seen as essential drivers in the propagation of malicious content [40,55] and cannot be targeted with persuasion strategies.

Network alteration strategies are standard practice in many social networks. Providers of social networks have been identifying, flagging, and removing malicious content and accounts on an ongoing basis [15,28]. However, despite the short-term effectiveness, bot removal has significant drawbacks. First, identifying and removing malicious agents requires constant attention and monitoring by social network providers

[19]. Since the deletion is transparent to the account owner, reentering the social network by creating new and more sophisticated malicious agents is difficult to restrict and monitor. Thus, identification and removal must be conducted on a rolling basis, rendering the strategy ineffective and expensive. Twitter has been forced to conduct regular bot removals [60], and Facebook regularly follows a defined set of actions to identify cyber threats on their platform [28]. Second, alteration strategies generally focus on alleviating the symptoms of manipulation in social networks rather than targeting the mechanisms such as manipulation exploits [48]. Third, content and bot removal have come under much scrutiny and fueled discussions on censorship and curtailing freedom of speech [10,28]. For example, given the ambiguity around fact-checking organizations, questions have been raised regarding bias while deeming specific content true or false [10].

We argue that given these limitations of both persuasion and the network alteration strategies, a superior network intervention strategy must meet the following requirements: (1) be able to target any user, i.e., private individuals, entities, and bots; (2) be sustainable in the long-term without the need for constant monitoring; (3) cater to the mechanisms of content propagation that malicious agents exploit; (4) avoid deletion of content to prevent accusations of censorship. In the following subsections, we present our proposed feed management strategy PDFM, its assumptions, and its properties.

3. Preferentiality dampened feed management

To combat malicious content in social networks, we propose a feed management strategy, PDFM, as an alternative network intervention strategy. This strategy aims to move away from the hard-handed and short-lived policy of content or bot removal to a more nuanced approach that targets content flagged as malicious and dampens the spread of this content. There are two core assertions upon which we build PDFM, i.e., (1) PA and (2) LIP. We first summarize and synthesize key literature on PA and LIP and then present the critical tenets of PDFM.

3.1. Preferential attachment

Research on PA revolves around the mechanisms that underlie formation, growth, and information diffusion in social networks. Barabasi and Albert [3], in their seminal work, theorize that a PA mechanism underlies the power-law distributions found in random networks (here, social networks). PA describes the tendency of new nodes to link to well-connected nodes rather than those poorly integrated with the rest of the network. PA is built upon the idea that individuals can use their social capital to accumulate even more social capital [41]. The logic follows the ‘rich get richer’ analogy, suggesting that those who start with more social capital will be more successful in accumulating even more of it [43]. Much prior research has demonstrated the role of PA in the formation of social networks such as Wikipedia, Twitter, and virtual graphical worlds [6,28,53]. New users are more likely to connect with the most popular members of a network instead of regular members of the network [28]. Social networks often build upon the mechanism to promote user engagement by making popular content readily available to new users, resulting in a network of few well-connected users and a long tail of poorly integrated users [6,53].

While the focus of PA has lain on the initial formation of networks, its role in content propagation has only been considered indirectly. The study by Lu et al. [41] represents one of the few endeavors connecting PA with content propagation. By investigating the emergence of opinion leadership in online review communities, Lu et al. [41] found PA to be one of the main drivers of new incoming links to that user. Lu et al. [41] further posit that by gaining popularity in a network, opinion leadership emerges through the increasing exposure of their content and its potential influence on other users in the network. Because of PA, once dominant reviewers established themselves in the network, it took much work for the new reviewers to become opinion leaders, irrespective of

quality and number of written reviews. Therefore, PA not only acts as the primary mechanism of network formation but also as a driver of content propagation in social networks.

3.2. Limited information processing

LIP is based on cognitive load theory that originates from cognitive psychology and explains how cognitive resources are used in learning processes and problem-solving [59]. Cognitive load theory describes the mental effort or the amount of working memory resources individuals use to perform tasks [29]. While some problems can be inherently cognitively taxing, cognitive load can also stem from factors surrounding a problem – i.e., extraneous cognitive load – for example, how a decision problem is instructed or presented. Cognitive load can be exceptionally high in online environments due to complexity, user familiarity, and website navigability [31,45]. Cognitive load is an essential determinant of user satisfaction [31]; overloading users’ cognitive resources in online environments has been associated with lower satisfaction [45].

Working memory capacity is limited. Prior research suggests that individuals can make a maximum of seven absolute judgments of unidimensional options [44]. In social networks, cognitive load is caused by various factors that tend to overcharge cognitive capacities, such as the amount of information, communication, system features, and social interaction [4,29]. Cognitive overload can further stress individuals’ LIP capacities. A piece of information is stored in working memory for a short 15 to 20-s time period, after which the information is either transferred into long-term memory or forgotten [51]. Once these capacities are exhausted, individuals often use workarounds such as relative judgment, heuristics, or social reference to solve decision problems [44]. Content propagation literature suggests that individuals’ LIP distinguishes the contagion of information in social networks from the contagion of, for instance, an infectious disease. While with a disease, the likelihood of infection increases with every additional exposure; in social networks, the more connected users are, the more likely they are to overlook a particular piece of information and thus escape infection [26].

Research on misinformation sharing has shown that respondents under high cognitive load (i.e., time pressure) are more likely to intuitively believe a fake headline than in a situation where they are given the time to deliberately reflect upon the veracity of the information [44,51]. Prior research has found that cognitive reasoning helped respondents discern fake from legitimate news headlines regardless of political opinions. This suggests that in situations where people do not use cognitive reasoning, for example, due to cognitive overload, people are more susceptible to believing misinformation [2,31,45,64].

While LIP plays a significant role concerning users’ cognitive processing capacities and attention span, and consequently, the content users consume and share in social networks, the activity of the PA mechanism further reinforces this effect. In line with prior research on cognitive load and the amount of information individuals can process (e.g., [26,44]), we argue that on the one hand, LIP causes individual users to only attend to the top n percent of their feed, on the other hand, PA generates a preferential rather than a random sorting of posts that will appear at the top of users’ content feed. The interplay of LIP and PA results in a disproportionate exposure of the most popular content by the most famous actors on the social network. In other words, popular nodes get the most attention under PA, but because human users are bound by their LIP, the content generated by less popular nodes gets less attention.

3.3. PDFM: Concepts and principles

The goal of PDFM as a feed management strategy is to curb the propagation of malicious content. Specifically, PDFM aims to neither remove the content nor the users that propagate the malicious content. Instead, the goal of PDFM is to deprioritize the flagged content. To

achieve this de-prioritization, we draw from the search result literature (see [68]). In doing so, we focus on two key content presentation processes: Selection and Sorting. Because of LIP, users tend to have a limited attention span [26], i.e., users only consume a subset or a slice of the entire content generated by social networks. This subset of the content is what we refer to as the users' content feed. Therefore, the content that gets 'selected' for the users' feed is a crucial characteristic that governs the content's overall exposure and engagement throughout the network.

The selection process may employ PA, such that the content gains a greater probability of getting selected for the content feed based on the content's existing popularity. Alternatively, content selection can be more arbitrary, such that all the content generated through the network has an equal chance of getting selected for the content feed. The latter scenario can be considered as a "random" selection approach. Such a "random" selection approach will likely dampen popular content's advantage over other non-popular content because the probability of being chosen is equal for every post.

On the other hand, the sorting process refers to the order of the selected content. Search advertising literature has provided evidence that results appearing on the top of a search list garner significantly more attention than those appearing later in the list. Following the same line of argument, the relative position of a post on the content feed plays a critical role in whether users are exposed to a particular content. This sorting process may also employ PA. Therefore, the content is not only selected based on its popularity but also ordered based on its relative popularity. Alternatively, the sorting process can employ organic criteria such as chronology or the user's relative proximity to the content source. Search marketing literature (e.g., [68]) demonstrates the efficacy of such organic listings compared to sponsored listings, similar to PA-promoted content in our context.

Depending upon the two processes—i.e., selection and sorting—we can have four different levels of PDFM—i.e., no PDFM, low PDFM, medium PDFM, and high PDFM (see Table 1). Recall that PDFM does not remove malicious content. Instead, PDFM dampens the popularity of malicious content with viral tendencies [64]. We argue that the nature of the selection and sorting processes determines the level of dampening caused by PDFM. More specifically, we argue that, on one extreme, no dampening occurs when the content feed employs PA-based selection and PA-based sorting. On the other extreme, high dampening occurs when the content feed employs non-PA-based selection and non-PA-based sorting. No PDFM provides the baseline to compare the efficacy of low PDFM (selection dampened), medium PDFM (sorting dampened), and high PDFM (both selection and sorting dampened).

To elucidate the distinctions in functionality and efficacy among the different levels of PDFM consider the hypothetical scenario of a social network user, referred to as user A. In this context, we next delve into the specific intricacies of the selection and sorting processes within the PDFM levels and highlight their significance. In the absence of PDFM, where both selection and sorting processes employ PA-based filtering, the posts appearing on user A's content feed are those generated by their most popular followees. These posts are subsequently ranked based on their popularity, culminating in a feed dominated by content from highly popular users. This configuration accentuates the amplification of

popular content, potentially leading to the reinforcement of certain viewpoints and trends.

Transitioning to the low PDFM level, the content selection process becomes non-PA-based, rendering each post equally probable for selection. However, the sorting process persists with PA-based ranking. Consequently, while the selection introduces randomness, the order in which posts appear still heavily favors popular content. This nuanced interplay between random selection and popularity-based sorting introduces an element of diversity in content while retaining the visibility of popular posts. In contrast, the medium PDFM level employs PA-based selection but embraces a non-PA-based sorting methodology, organizing posts chronologically rather than by popularity. This divergence in the sorting approach disrupts the traditional hierarchy of popular content dominance. As a result, while popular posts continue to be selected more frequently due to the PA-based selection, their positioning in the feed is dictated by their timing of creation, promoting a more balanced exposure across all content sources.

The rationale behind the emphasis on the selection and sorting processes lies in their role in shaping the content distribution landscape. Selection determines which posts enter the user's feed, and sorting dictates the order in which they are presented. By altering these processes, PDFM levels can influence the prominence and visibility of different types of content. A focus on the chronological sorting aspect within medium PDFM, for instance, aims to reduce the undue amplification of popular content and instead encourage a more even representation of content sources, thereby fostering a more diverse and balanced content experience for the user. Ultimately, the delicate interplay between selection and sorting processes within PDFM levels introduces strategic adjustments that impact the way users engage with content. By strategically tailoring the impact of these mechanisms, PDFM seeks to achieve its overarching goals of counteracting undue content amplification, enhancing content diversity, and providing a more equitable content exposure for users like A within the complex socio-technical dynamics of social networks.

4. Methodology

We draw on complex adaptive systems to build an ABM which allows us to simulate propagation patterns of malicious content in the Twitter network. The ABM simulation of the Twitter network provides a useful test environment for evaluating different network intervention strategies to curb the propagation of malicious content. In simulation experiments, we test bot removal and PDFM at different levels to evaluate their efficacy to dampen malicious content propagation. In the following subsections, we will briefly introduce complex adaptive systems and ABM, explain our detailed ABM setup, the experimental design as well as conditions and measures used, and finally discuss the validation of our ABM.

4.1. Agent-based models

Complex adaptive systems are "systems composed of interacting agents described in terms of rules. The agents adapt by changing their rules as experience accumulates" [27, p. 1]. CAS demonstrate evolution through the dynamic and adaptive interactions between various agents and are characterized by self-organization and emergence [65]. ABM represents a bottom-up computational modeling approach that can be used to model and study CAS [46]. The simulated individual agents with distinct attributes and behaviors based on predefined rules produce emergent macro-level patterns. ABM are particularly suited for analyzing multi-level phenomena involving learning agents and self-organizing interactions that are challenging to model with differential Eqs. [52]. ABM has gained traction in management and IS research to computationally model and simulate a variety of processes in complex systems [e.g., 22, 47] and have been deemed specifically suitable to study social networks. Recent work has employed ABMs to study such

Table 1
Presence of PA across the different levels of PDFM.

Process step	PA in the status quo (i.e., no PDFM)	PA under PDFM		
		Low PDFM	Medium PDFM	High PDFM
Content selection	✓	X	✓	X
Content sorting	✓	✓	X	X

Notes: PA: Preferential Attachment; PDFM: Preferentiality Dampened Feed Management

aspects of social networks as the impact of manipulated opinions [55], and rumor propagation [39]. ABM is a proven tool for experimentation in multi-level contexts like social networks [22,55] and ABM provide a tool to model adaptive behaviors by heterogeneous, learning agents and is thus a suitable approach to study the complex phenomenon of content propagation.

To develop our ABM, we chose a network-based topology to form agent neighborhoods based on links [65]. Network-based typologies align well with empirical social network communities [33]. Moreover, they allow for accommodating complex interactions within high-dimensional topologies of social networks [55]. Thus, we derived the underlying agent ecosystem from prior literature. Our simulation model involves eight agent classes, each with distinct attributes (prevalence, attention span, followers) and behaviors (following, posting, reposting). The classes are: simple bot, sophisticated bot, influencer, celebrity, broadcaster, amplifier, commentator, and viewer. This agent ecosystem, validated using empirical data from campaigns, forms the foundation of our ABM.

Based on our classification of agents and their characteristics, we constructed a dynamic agent-based ecosystem and define specific interactions within it. Two central mechanisms are essential in this construction and are integrated into the ABM at different points. PA is implemented in various phases. Initially, during the network's inception phase, PA is utilized to establish follower-followee links. Subsequently, PA extends its influence on content dissemination, shaping the users' content feed and thereby determining the visibility and spread of information within the network. Consequently, PA has a direct impact on user engagement with the content. In parallel, the LIP capacity of agents dictates the volume of content they engage with before reaching cognitive fatigue. While LIP is incorporated at the stage of individual agent behavior, it nonetheless has a substantial impact on user-content engagement, contributing to the system's emergent properties. Modeling these mechanisms requires meticulous attention due to their potential widespread effects on user interactions and the overall emergent behavior of the system.

We develop our ABM in Python, using object-oriented programming to simulate the social network that underpins agent interactions and communications. During the network generation phase, agents select whom to follow based on a predefined range that is specific to their class. We draw on established literature, notably the work of Barabasi [3] and Ross et al. [55], to inform our use of the PA mechanism. This not only shapes initial follower connections but also profoundly affects agents' subsequent behaviors, like reading and sharing content. As a result, our network exhibits a scale-free degree distribution, which means that some agents have many more followers than others—a pattern that mirrors the uneven follower distributions seen in actual social networks. This feature of our model captures the natural diversity in user connections, adding a layer of realism by reflecting the inherent variations found in social media platforms.

In our simulation, experiment conditions are designed to assess the impact of a specific level of intervention. In the first experiment, we examine the effects of bot removal in four experimental conditions, this includes the baseline scenario for comparative purposes. In the second experiment, we examine the effects of different levels of PDFM in three experimental conditions against a similar baseline condition. To ensure robustness and reliability of our findings, every experimental condition is subjected to ten iterations, culminating in a comprehensive total of ninety trials within the entire simulation framework.

Each simulation runs for 30 days, and we model each day as a 'tick', resulting in a total of 30 ticks. The simulation incorporates 10,000 agents to emulate an active, scaled-down version of a Twitter-like social network. By default, posts are organized using a 'TWTD' (Default) sorting rule, which refers to the posts appearing in a chronological manner, so the latest posts appear first in the feed of users of the social network. Additionally, we have incorporated alternative mechanisms such as 'RTWC' (Retweet Count) and 'RTWD' (Retweet with Decay) to

replicate different patterns of information spread. RTWC refers to the scenario where posts with highest repost counts appear at the top of the feed. In RTWD the order of the posts appearing on the users' feed is a combination of the reposts count as well the time they were generated. Additionally, the agents' engagement with tweets is governed by a default 'RNDM' (Random) picking policy, which stands in contrast to the 'PFAT' (Preferential Attachment) method, a strategy intended to mimic diverse attention distributions. We discuss this according the four levels of PDFM in section 5.2.

To ensure proportional scaling of certain behaviors in simulations where the agent tally is below our baseline, we set a baseline tally at 10,000 agents. Each of our eight agent classes plays a unique role within the network and exhibits specific behaviors, including how often they tweet, their engagement levels, and the number of followees they can have, which varies from low to high depending on their class. Furthermore, follower connections are established in a non-deterministic fashion. Each agent is programmed to follow others based on a range determined by their 'followee mark'—a measure of their propensity to follow—and adjusted according to the baseline factor. This approach not only brings variability into our simulation but also aligns with the scale-free network characteristics found in real-world social media platforms, where connectivity is not uniformly distributed across all users.

4.2. Experiment design

We conduct two simulation experiments to evaluate the effectiveness of different network intervention strategies—bot removal and PDFM—on mitigating the spread of malicious content. The first experiment focuses on the effectiveness of eliminating malicious bot accounts. For this, we compare the effectiveness of account removal at four different levels. We use four different manipulations, namely removing 25%, 50%, 75%, and 99% of bots and compare these to a no-intervention baseline simulation to find the optimal level of account removal. The second focuses on the effectiveness of PDFM. For this, we compare the effectiveness of PDFM at four different levels. Like in the first experiment, we use four different manipulations, namely No PDFM, Low PDFM, Medium PDFM, and High PDFM and compare these to a no-intervention baseline simulation to find the optimal level of PDFM.

To assess the comparative effectiveness of network interventions, we use two measures for content propagation: engagement and exposure. Engagement is operationalized as the repost distribution, that is, counting reposts for each generated post. This metric of content propagation has been widely used in prior research [17,64]. Additionally, exposure is operationalized as the post readership, that is, tracking the number of users viewing each post within their feed. We considered this metric essential for understanding content spread. While repost distribution is accessible through popular social networking platforms, capturing readership distribution requires a simulation model like the one presented here.

4.3. Model validation

We performed internal and external validation of our ABM of the Twitter network [22] drawing on Ross et al. [55] who argue model validation is vital to ensure it behaves similarly to a real-life social network agent. Internal validation ensures that the model's inner workings are consistent and that the components of the model behave as expected. External validation introduces new data previously unseen by the model and checks whether the model still performs consistently with the expectations. We followed the steps by Ross et al. [55] to ascertain internal validation and determine the amount of stochastic variability throughout the network simulations. In addition, we examined the emergent agent behavior and compared it with the patterns in previous literature [63]. We found that both stochastic variabilities and the agent behavior consistency indicated internal validity. These internal validation checks are consistent with ABM studies in prior literature (for

example, see [22]).

To validate our model internally, we analyzed data from 28 malicious campaigns to develop a user typology, reserving data from 9 additional campaigns to validate our findings, adhering to a standard 75–25 training-test split. Particularly, we focused on three campaigns originating from Russia, consisting of more than 70,000 posts, to categorize users by agent type. We then systematically removed different agent types from these campaigns, observing the effect on content spread. These steps were replicated in our ABM, where the removal of specific agent types similarly resulted in a decline in content propagation, mirroring the real-world data. This validation confirms our ABM's accuracy in simulating the dynamics of actual malicious campaigns.

To conduct our external model validation, we created a customized agent called 'CELEBRITY' in our network. This agent mimics the behavior of a real-life celebrity account. Our choice of a celebrity agent is grounded in the thought that due to their wider reach throughout the network, changes in their behavior have a ripple effect throughout the network and, therefore, help us determine whether our model emulates real-life social networks. We chose the real-life account of former US president Donald Trump for our data collection because this account was the most prominent celebrity account on Twitter at the time and was affected significantly by the removal of bots by Twitter [28]. To validate our model, we monitored changes in CELEBRITY account statistics in real life and compared them with that of the customized agent in the simulation. As it is unknown how many accounts in the Twitter network were truly bots before and after the bot removal exercise, we ran simulations at different levels of bot removal. Specifically, we ran simulations at 25, 50, 75, and 99% of bot removal. To establish whether the simulation agent is emulating real-life accounts, we needed to find a measure that could capture the propagation of the posts. As such, we focused on the varying amplitudes of repost levels for over 300 real-life tweets and compared them to the repost levels of the simulated posts in the ABM.

We found that for our real-life CELEBRITY agent, the average drop in amplitudes of repost count received per post from before bot removal (June 15–July 30, 2018) to after bot removal (November 1–16, 2018) was 56.8%. Comparing this to our simulation agent, we found that the drop in amplitudes of repost count, on average, accounts for 42% (for the removal of 25% of the existing bot population), 54% (for 50% bot removal), 63% (for 75% bot removal), and nearly 70% (for 99% bot removal). This shows that the 50% bot removal strategy leads to a similar drop in repost amplitude levels compared to real-life posts. Other measures such as repost count, maximum repost count, minimum repost count, mean repost, repost standard deviation, and repost median also distinguish the simulated bot removal at 50% and the real-life actor we used in our study.

We conducted two separate Welch *t*-tests to conduct our validation—one for internal validation and the other for external validation. For this validation, we randomly selected over 4400 posts from our dataset consisting of over 70,000 real tweets originating from Russian and Celebrity malicious campaigns. We tested for propagation before and after the account removal and then tested for the differences in means controlling for the size of the network. We found that there were no significant differences in propagation of posts between the real dataset and our simulated observations ($t = 1.26$; $p = 0.21$) providing evidence for internal validity. For external validation, we again found no significant differences in propagation of posts between the real dataset and our simulated observations ($t = 1.52$; $p = 0.16$).

5. Results

5.1. Bot removal

In the first experiment, we investigated the effectiveness of network alteration strategies that profess to remove bots propagating malicious content. In doing so, we analyzed the impact of bot removal at four

levels. The removal of bots had a noteworthy impact on the number of reposts garnered by users whose posts are usually shared in high numbers. As expected, we observed a downward trend as more bots were removed. On average, at 99% bot removal, the broadcast and celebrity repost counts were reduced by 70 %, and the influencers lost 50 % of their repost count. This makes intuitive sense because bots play a significant role in propagation based on reposting actions.

Bot removal significantly impacted the number of followers for each kind of agent in our model. We observed that the number of followers decreased for every agent kind. Unlike more popular agent types such as influencers, celebrities, and broadcasters, bot removal had a limited impact on amplifier and commentator agent reposts. These agents received fewer reposts as compared to other agent accounts. Furthermore, we observed that the bot removal did not significantly impact an agent's reposting activity, which is unaffected by the removal of other accounts. From the values provided below in Tables 2 and 3, we can observe that mean values for both readership and reposting increase as bots are removed, indicating that on average, posts are read and shared by more users. The variance in both distributions also rises, suggesting a broader spread of engagement levels across posts. Notably, the skewness and kurtosis values decrease in all scenarios as bots are progressively removed. This decrease in skewness points to a lessening of asymmetry in the distributions, with fewer posts dominating the extreme end of engagement. Similarly, the decline in kurtosis signifies that the heavy tails of the distribution become less pronounced, indicating that exceptionally high engagement numbers become rarer and less extreme.

We can infer from these observations that bot activity on the network disproportionately inflates the engagement metrics of certain posts, leading to highly skewed distributions. This implies that a small number of posts may be receiving a large number of retweets or readership counts due to bot amplification rather than organic user engagement. The increased mean and variance after bot removal suggest that genuine engagement is more evenly distributed among posts, reflecting a broader interest across the network's content. While a 99% bot removal rate yields the most optimal outcomes, as underscored in our discussion, it also requires substantial resources and effort. Notably, Table 2 reveals that a 50% bot removal strategy leads to a significant and more evenly distributed increase in network readership. On the other hand, Table 3 indicates that both 75% and 99% bot removal rates exert a more pronounced effect on reposting distribution, suggesting that engagement metrics respond differently to bot removal interventions. This differentiation implies that substantial improvements in network engagement can be achieved with a 50% bot removal rate, which presents a more cost-effective and attainable objective than striving for a 99% removal rate.

We also captured the total number of posts generated, the total number of reposts, and the total number of reads received by these posts throughout the simulation. We observed that bot removal did curb the spread of targeted malicious content but also significantly reduced the generation and spread of non-malicious content in the network. The total number of posts generated was reduced by 80 %, and the corresponding drop in post reads equaled nearly 40 %.

5.2. Preferentiality dampened feed management

Experiment 2 investigated how PDFM dampened the reach of

Table 2
Statistics for Readership Distribution for Scenarios.

Scenario	Mean	Variance	Skew	Kurtosis
No Intervention	8.67	1378.71	10.93	128.7
After 25% Bot Removal	10.76	1459.88	9.5	98.7
After 50% Bot Removal	14.08	1854.69	7.86	77.89
After 75% Bot Removal	16.76	2355.92	6.39	44.42
After 99% Bot Removal	22.61	4924.71	3.95	15.38

Table 3
Statistics for Reposting Distribution for Scenarios.

Scenario	Mean	Variance	Skew	Kurtosis
No Intervention	0.96	128.06	16.93	324.54
After 25% Bot Removal	1.03	109.33	15.65	272.89
After 50% Bot Removal	1.21	117.96	13.41	241.64
After 75% Bot Removal	1.48	120.68	10.78	133.39
After 99% Bot Removal	2.18	144.83	7.09	55.28

malicious influential accounts, thereby limiting the propagation of harmful content throughout the network. We compared four conditions: (1) no PDFM in which users' content feeds are preferentially sorted (as is the common practice in most feed-based social networks), (2) low PDFM in which PA is removed from content selection to a feed (3) medium PDFM in which PA is removed from content sorting to a feed and (4) high PDFM in which PA is eliminated and content is selected randomly and sorted chronologically.

To compare the interventions, we captured both the readership and repost distributions of the posts during the simulations. Given the popularity of the content generated by influencers or celebrities, the repost distributions likely reflect high levels of variance, skew, and kurtosis. If the interventions were successful, the readership and repost distributions should “flatten”—i.e., become less skewed and have more reads and reposts for a broader range of posts.

We observed that the readership is intensely concentrated on a few selective posts when No PDFM is employed. Most popular posts were read by more than 50 % of all users in the network. In contrast, the intervention of High PDFM showed that the readership and reposting distributions were much more spread out. For example, for High PDFM intervention, the most popular post was read by less than 20 % of the network population, a more than 60 % drop from the readership of the most popular post under the No PDFM intervention scenario. As demonstrated in Table 4, the kurtosis of the readership distribution dropped from 108.33 under the No PDFM set up to 11.75 when neither sorting nor selection of posts employed PA (High PDFM). The corresponding drop in skew was from 10.11 to 2.57, leading to our assertion that the number of reads was no longer dominated by a select few posts from influential users. We also observed that dampening the PA in the sorting process (Medium PDFM) had a more significant impact on flattening the curve than dampening the PA in the selection process (Low PDFM). Across all scenarios, we observe a gradual increase in the mean readership and reposting. A rise in the means signifies that the content is disseminated more broadly across the network, resulting in extended reach of a greater number of posts as PA is gradually removed. This suggests that a more extensive and diverse audience is exposed to the shared content, thereby cultivating a more diverse information environment.

Content propagation was more evenly distributed when PA was not applied. When No PDFM is employed, reposts were concentrated for the most popular posts only, but dampening PA demonstrates a more uniform distribution of reposts across the network. Again, there is evidence of a significant difference in the exposure that a post garnered throughout the network, with or without PA as part of feed management. We can observe from Table 5 that the drop-in kurtosis for the repost distribution from the No PDFM setup to the High PDFM setup was 415.78 to 12.73, respectively. The drop in skewness was 19.46 to 2.75, meaning that reposts in the network were no longer being dominated by

Table 4
Statistics for Readership Distributions.

Scenario	Mean	Variance	Skew	Kurtosis
No PDFM	6.9	2134.5	10.11	108.33
Low PDFM	7.02	1630.98	9.07	97.9
Medium PDFM	28.03	657.32	3.6	21.8
High PDFM	28.44	600.41	2.57	11.75

Table 5
Statistics for Reposting Distributions.

Scenario	Mean	Variance	Skew	Kurtosis
No PDFM	0.89	200.32	19.46	415.78
Low PDFM	1.35	96.8	12.1	165.44
Medium PDFM	2.01	23.93	6.68	59.65
High PDFM	2.38	9.97	2.75	12.73

a select few posts.

The rise in means observed in the Tables 4 and 5 carries several significant implications. Increase in means underscores the broader dissemination of content across the platform, as evident in the accompanying graphs displaying the extended range of posts involved in content propagation. This suggests that a more extensive and diverse audience is exposed to the shared content, thereby cultivating a more dynamic information environment. Users may be motivated to interact with content on the platform either because they seek confirmation of their existing viewpoints (content they like) or because they crave diverse and contrasting content that challenges their current perspectives [2,18,32]. This understanding could inform the platform's approach to content curation, aiming to satisfy the different content consumption preferences that exist within its user community.

The statistical significance was tested by comparing the means, variance, skew, and kurtosis across multiple simulation runs. Using the average results from ten simulation runs for each intervention and Welch's *t*-test, we conclude that there is a statistically significant difference between all measures among interventions employed with and without PA at a significance level of 0.01. The goal of PDFM, consisting of four different scenarios, was to find a balance that prevents the winner-take-all outcomes often produced by PA. A more balanced content feed management policy promotes popular content while blending in other content. When compared to bot removal strategies, we observe that dialing down on PA significantly impacts skew and kurtosis of the readership and reposting distributions. We also observe from overall network statistics that even though strategies of dialing down PA reduced the influence of the select few accounts, network traffic was not hampered, and the total number of posts, reposts, and reads remained consistent and, in some cases, even increased. Thus, PDFM did not reduce the overall network traffic, a metric significantly important for network providers. Overall, for flagged malicious accounts, when PA was removed (High PDFM), readership decreased by nearly 350%, and reposting decreased by 300% compared to when PA was employed (No PDFM). Therefore, both High PDFM and Medium PDFM are extremely effective strategies against malicious content propagation. But High PDFM strategy implementation may be a concern for the network providers similar to bot removal as it will be close to absolute censorship and may lead to malicious agents noticing and opening newer accounts leading to same issues that are associated with bot removal strategy. However, Medium PDFM may be a better alternative as even though it reduces malicious propagation at par with high levels of bot removal, it is not as restrictive or evident as High PDFM.

5.3. Comparing bot removal and PDFM

Comparing the results from both sets of experiments, we can observe that both bot removal and PDFM strategies are effective at reducing the propagation of malicious content. As for bot removal, despite its effectiveness in increasing mean readership and reposting and reducing distribution skewness and kurtosis, it is subject to a substantial drawback. Our results indicate that bot removal significantly reduces network traffic. This not only affects malicious agents in the network but also legitimate agents posting legitimate content. Furthermore, removing bots is an extremely laborious and complex endeavor, mainly due to the difficulties in accurately detecting bots. Our analysis indicates that Twitter's 2018 network alteration removed roughly 50% of the bots

from the network. Comparing the propagation outcomes of bot removal and PDFM, it becomes clear that Low PDFM is able to achieve dampening results that lie between those observed by 75% and 99% bot removal. While Medium and High PDFM achieve even better effectiveness in terms of dampening malicious content propagation, complete deprioritization as in High PDFM (i.e., random selection and chronological sorting) might not be consistent with social networks' user engagement-based revenue models.

6. Discussion

This research aimed to analyze different network interventions designed to curb the spread of malicious content on social networks. We proposed PDFM based on content propagation literature that highlighted the need for accounting for the role of the two mechanisms PA and LIP in the propagation of content on social networks. PDFM demonstrates that PA is crucial during initial network formation and plays a key role in how content propagates throughout the network. Similarly, we were able to showcase the role of LIP in content consumption on social networks via curated content feeds in conjunction with PA. Using ABM, we were able to emulate a social network. Using data from Twitter, we validated our proposed model. We found support for PDFM performing better than the established interventions in literature and practice using metrics developed to capture the engagement and exposure of content on social networks.

6.1. Contributions to scholarship

Our study makes three key contributions to scholarship. First, we contribute to the network intervention literature by proposing and substantiating the effectiveness of a new class of network intervention strategies to curb the spread of malicious content in social networks. Prior research has brought forward myriad interventions that can be categorized as persuasion or network alteration strategies [62]. In proposing PDFM, we extend the network intervention literature by an alternate class of strategies, namely, feed management. Unlike many established strategies (e.g., content or bot removal) [62], PDFM draws upon the underlying mechanisms of malicious content propagation (i.e., PA and LIP) rather than focusing on dealing with malicious content or agents themselves. Thus, PDFM responds to the critique of network alteration strategies being effortful, short-lived, and restricting freedom of speech. Indeed, our results indicate that increasing levels of PDFM correspond to a more uniform distribution of content readership throughout the network. Moreover, PDFM has the capacity to target synthetic agents in the social network responsible for spreading malicious content, unlike persuasion strategies that include nudges, misinformation flagging, or fact-checking alerts targeting human users [14,20,57]. Hence, PDFM expands the solution space in the network intervention literature by adding a new class of strategies that neither target the nodes nor the links between them [17]. Rather than targeting users or content directly, PDFM considers the dynamic interplay between the users of the network (i.e., posting, viewing, and reposting content) and the algorithmic filtering (i.e., preferential selection and sorting of content) that give rise to the specific patterns of malicious content propagation. In sum, PDFM tackles the mechanisms of content propagation exploited by malicious agents instead of treating the symptoms.

Second, we contribute to the social network literature by substantiating the roles of PA and LIP and the importance of considering them as mechanisms in scholarly research on malicious behavior in social networks. While PA has been recognized as a critical mechanism in the formation of social networks (e.g., [32,44]), its role in the propagation of content has only marginally been discerned [41]. Nevertheless, our findings lend support to the theorizing that PA also acts as a mechanism that governs content propagation [58]. Applying PDFM has a dual impact on the viewership of content. On the one hand, malicious content

is dampened from users' content feeds and thus viewed less. On the other hand, due to the increased exposure across content feeds, legitimate content is viewed more, independent of its popularity. While this is an essential outcome of fighting misinformation, it is also interesting from a content consumption perspective. As PDFM leads to an equal distribution of viewership across non-malicious content, as a second-order effect, it might positively affect information source diversity in social networks (cf. [35]) and moderate the opinion polarization at the core of social networks, which is caused mainly by popular nodes [58]. In other words, PDFM may reduce the risk of echo chambers [58] and the political slant in content consumption and propagation [58]. Our findings further corroborate the central role of human LIP capacities in content propagation in social networks and thus help explain content virality. While cognitive load is known to be exceptionally high on social networks [4,29,31,45] and tends to play an essential part in the diffusion of false information [2,64], most network alteration literature has given little attention to the human property of LIP (e.g., [26]). Our research incorporates LIP into the heuristic decision-making for content consumption. Specifically, by incorporating attention span to curate content feeds, our work explains the impact of LIP on content consumption by the users of social networks. Our study's results support the assertion that human users only consume the top portion of their content feed, explaining why specific viral (especially malicious) content may spread faster and broadly across networks than legitimate content [64]. Therefore, our findings offer an explanation for previously observed but poorly understood phenomena, such as the variation in the spreading speed of fake and legitimate news [64].

Third, our work contributes to the ABM methodology. ABM has recently been used to study complexity in networks in domains such as finance [8], medicine [11], and computer science [17]. We extend the application of ABM methodology toward its use as a social network policy evaluation tool. Closeness to reality has been a frequent point of criticism of ABMs [11,22,55]. Our specific approach to ABM caters to this critique in two ways. On the one hand, we ascertain that our model emulates a real-world social network (i.e., Twitter) by incorporating empirically grounded formation and propagation mechanisms such as PA, realistic user properties (e.g., LIP), and a realistic network structure (i.e., agent typology). On the other hand, we employ an innovative validation technique of using data from Twitter's real-life bot removal, contrasting the real-life and simulated propagation of content using the 'CELEBRITY' agent. Hence, our model provides a valuable basis to inform future simulations of social networks and to study the nuanced impact of different policies or network interventions. The nuanced impact of policies on network traffic and the subsequent composition of the social network can only be effectively studied in a simulated environment, such as the one presented in our work because simulated environments can control for any extraneous variables. The ABM model we developed has been made available for public use on GitHub and can be used by future researchers to conduct future investigations. Besides the model, researchers using ABM for social network-related inquiries may draw on our validation approach and the specific metrics we developed to capture content propagation, namely, readership and repost distribution.

6.2. Practical implications

Our findings have three critical practical implications. First, targeting social network providers, when employing mostly ad-based revenue models that follow a pay-per-click or pay-per-view logic, social networks tend to engage and retain users to have a vivid and interactive platform. Therefore, information pollution through malicious agents negatively affects the users and "paying customers" (i.e., entities placing ads) of a social network, causing potential misrepresentation of legitimate information and reputational damages for the platform. Hence, providers of social networks have a strong interest in preventing malicious agents from polluting the information on the platform, all the while retaining

users. Applying PDFM allows social networks to inhibit malicious content more efficiently and sustainably than the widely used bot removal strategy. By attenuating the PA mechanism in the content sorting, the exposure of malicious agents is automatically reduced without requiring working power to manually and repeatedly review, identify, and remove malicious content and accounts, which can be costly for platform providers. Moreover, social networks can benefit from using simulation tools to predict possible trends of malicious behavior and experiment with different mitigation strategies and nuances thereof. For this purpose, they may draw on the proposed ABM built upon established mechanisms (PA), agents (see our typology), and realistic agent characteristics.

Second, for the benefit of the users, PDFM enhances user experience on social networks. Non-malicious users have two goals while using online social networks [8]. First, users treat social networks as news sources that will supply them with legitimate content and information for opinion formation [8]. Naturally, users seek autonomy and want to avoid being manipulated. Second, social networks are employed for self-presentational purposes and to create outreach [64]. To uphold one's public image and reputation, it is vital to distribute legitimate information that reflects and strengthens personal stances and opinions and not so-called 'fake news.' By dampening the possibilities for manipulation and information pollution, PDFM serves users' best interests in allowing them to use online social networks without fearing manipulation and being abused by malicious agents to propagate their malicious content.

Finally, our findings have implications for the overall society. Online social networks not only serve platform providers and users in pursuing their economic and personal goals but are – as increasingly influential media for news consumption and opinion formation – of public interest. The possibility to gather information and form an unbiased opinion represents a fundamental tenet of our democratic process. Social networks are mostly “outlaw territory,” with little to no regulation (through providers). PDFM supports the maintenance of an unbiased opinion climate and the reduction of misinformation and manipulation in a non-invasive way by neutralizing the exposure of malicious agents and thereby reducing the predominance of malicious content [64]. If PDFM were to be applied in practice, the legitimacy of content in social networks would likely be enhanced. Thus, the landscape of legitimate media for information search and public opinion formation can be broadened.

7. Limitations and further research

Like any research, our work has limitations. The complexity of the simulated social network model we employed is limited and does not consider all possible ways malicious content can propagate on social networks. For example, the study does not simulate any use of graphical and textual content of malicious messages. Further research should expand on these dynamic network structures and user roles to explore the efficacy of different classes of network interventions. The static nature of the network in our current analysis also is a limitation. Social networks are dynamic entities. While this study deliberately constrains the network to a static state to isolate and examine the effects of PA and LIP on content propagation, incorporating these dynamics in the future will enrich the model's realism and relevance. The network size is currently set at ten thousand agents. Future research should examine if adding more agents may help uncover more nuanced mechanisms that can help develop more nuanced policies to mitigate malicious content online. Future research can use the ABM presented in this paper and propose other alternative strategies that media networks can adapt. While our findings may generalize to feed-based social networks, they do not apply to search-based social networks (e.g., Reddit). Future research should also diversify the heterogeneity of defined agents to reflect ongoing changes in the environment of social networks. Finally, we recognize that our research is limited in its scope to develop and

evaluate the effectiveness of network interventions that help address practical problems such as cost-effectiveness, long-term sustainability, and ethical considerations regarding freedom of speech. While our findings show that PDFM successfully dampens the spread of malicious content, particular interests of platform providers with respect to user engagement, user retention, source diversity, and revenue streams should be considered in future research.

8. Conclusions

Mitigating malicious content propagation online has become one of the biggest challenges of the modern era. Network alteration interventions have long been the only means to curb malicious content. This research demonstrated the effectiveness of an alternative strategy—PDFM—that employs dampening of PA and considers LIP to curb malicious content effectively. PA and LIP have been studied in the context of social networks but not from the lens of content propagation. Using ABM, we provide a nuanced view of how PDFM can curb malicious content propagation without raising issues such as censorship or compromising network traffic and thus platform revenues. Our approach provides insights into the influence of feed curation on content consumption. Our results have vital implications for social network providers, their users, and society. PDFM does not require manual and repeated identification or removal of accounts. By dampening certain content that may indulge in manipulation and misinformation, PDFM serves as an alternative strategy that can be more effective in curbing malicious content.

CRedit authorship contribution statement

Saurav Chakraborty: Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation, Conceptualization. **Sandeep Goyal:** Writing – review & editing, Writing – original draft. **Annamina Rieder:** Writing – review & editing, Writing – original draft, Conceptualization. **Agnieszka Onuchowska:** Writing – original draft, Investigation, Conceptualization. **Donald J. Berndt:** Methodology, Data curation, Conceptualization.

Declaration of competing interest

All authors certify that they do not have any financial or personal relationships that may be perceived as influencing their work.

Data availability

The authors do not have permission to share data.

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